AI Research Copilot — Executive Summary & Project Contribution

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1. Core Problem Statement

Retrieval-augmented scientific question-answering systems must decide whether retrieved passages contain enough evidence to answer a question. High relevance scores and repeated passages do not necessarily establish that all requested information is available. This project investigates whether exact deduplication and paper/section features improve evidence-sufficiency ranking over relevance-only baselines, while measuring acceptance coverage and the proportion of accepted contexts judged insufficient.

2. System Architecture vs. Experimental Scope

The software deliverable implements an end-to-end scientific literature assistant: Docling/pypdf ingestion with section regex parsing, ChromaDB dense retrieval (nomic-embed-text), BM25 sparse keyword search, BAAI/bge-reranker-base cross-encoding, a NetworkX citation graph with PyVis interactive visualization, and a Streamlit UI. However, the scientific experiment strictly evaluates the selective evidence-sufficiency gating mechanism on frozen context records; no unverified claims of end-to-end generation accuracy are made.

3. Controlled Benchmark & Methodology

We built a 120-family scientific benchmark across 160 papers (quarantining 40 historical development papers), split into 60 train, 24 validation, and 36 test families (144 test contexts across 36 paper clusters). Each family features 4 conditions: supported_candidate (4 passages), relevant_replacement (4 passages), duplicate_replacement (4 passages), and natural_retrieval (up to 6 passages within 2,000 tokens). Models (B0 Always Accept, B1 Max Relevance, B2 Historical Gate, M1 Relevance Raw, M2 Relevance Dedup, M3 Full Raw, M4 Full Dedup, and Length Diagnostic) were fit on train, tuned and frozen on validation, and evaluated on test with zero fitting. Models and thresholds were completely frozen before the new v2 test construction, while disclosing historical test-paper exposure during an earlier invalid Stage 4 run. Uncertainty was evaluated via 2,000 paired cluster bootstrap resamples (seed 20261104, clustered by primary paper ID).

4. Key Empirical Findings

• **Primary Hypothesis (M4 < M1):** On 108 controlled test contexts, M4 achieved discrete AURC = 0.4400 vs. M1 = 0.4329 (Delta = +0.0071, 95% CI [-0.0503, +0.0679]). The hypothesized improvement is NOT supported. The interval includes differences favoring either method; this study does not establish equivalence. Baseline B1 (Max Relevance) achieved the lowest observed controlled-context AURC (0.4049). The Length Diagnostic achieved discrete AURC = 0.6798, establishing only that this fitted diagnostic performed poorly.

• **Descriptive Multi-Passage Subgroup:** On 11 multi-passage families (33 controlled contexts), deduplicated models exhibited lower absolute discrete AURC (M2=0.3508, M4=0.3803) than raw models (M1=0.6894, M3=0.6900). This is a descriptive subgroup observation and does not establish causal claims.

• **Natural Retrieval Generalization:** On 36 natural contexts, M4 achieved the lowest observed AURC (0.0501). However, with only 3 insufficient contexts, the 95% CI spans zero ([-0.1176, +0.0160]).

• **Duplicate Invariance Property:** Under explicit operating assumptions, M2 and M4 confer bounded duplicate invariance (observed 0.000000 score change under post-selection duplicate injection on fixed contexts), while raw models suffer >0.31 score distortion.

5. Disclosures & Academic Rigor

Models and thresholds were frozen prior to new v2 test construction, with full disclosure of historical test-paper exposure during an earlier invalid Stage 4 run. Test labels were adjudicated via an untruncated AI assistant reading protocol (196 verified quotes), not independent human gold raters. Controlled contexts have high synthetic insufficiency (66.7%) compared to natural retrieval (8.3%). Commercial API spend was strictly $0.00.